

JAIKRIK ‘JJ’ MCCAULEY

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Summary

Full-Stack Software Engineer with experience taking production systems from initial architecture through cloud deployment. Currently serving as lead developer under contract for Tenaxiom Tech, building a multi-tenant Django and React facility management platform with PostgreSQL Row-Level Security and automated testing pipelines. Previously built real-time analytics dashboards for federal clients that filtered over 99.9% of network traffic, fine-tuned LLMs, and engineered vision pipelines using YOLO and OpenCV. Seeking to leverage this full-stack ownership and system design expertise to deliver high-impact software products.

Education

Salisbury University

Aug 2022 - May 2026

Bachelor of Science, Computer Science, Data Science (GPA: 3.85)

Salisbury, MD

• **Achievements:** Magna Cum Laude, Honors Graduate, Distinguished Computer Scientist Award, UPE Academic Achievement Award, Upsilon Pi Epsilon (UPE) President, Pi Mu Epsilon (PME) Inductee, Chi Alpha Sigma Inductee, Computer Science, Math, & Data Science Club President, CSC Academic All-District Team

Experience

Tenaxiom Tech | Lead Software Developer

Jun 2026 - Present

- Led software development for a gov-tech startup, owning the technical architecture, development, and production launch of a regulated operations platform and client portal on AWS
- Architected a Django and PostgreSQL backend with Row-Level Security for multi-tenant data isolation, server-authored audit logging, and an automated pipeline that generates type-safe TypeScript clients from OpenAPI schemas, which reduced client-side integration time and ensured data security across tenants
- Implemented enterprise authentication using AWS Cognito OIDC with Fernet-encrypted session tokens, tenant-scoped access controls, and security audit logs on AWS RDS with strict SSL verification, resulting in compliant, secure user access for all tenants
- Engineered a real-time compliance engine that projects statutory SLA deadlines and working-day review timers across workflows, and built an automated audit-readiness dashboard that scores compliance for active cases, helping the team meet regulatory deadlines consistently
- Deployed the cloud infrastructure on AWS using Docker, S3, Redis, and Celery, while maintaining a 90%+ CI test coverage floor with Vitest, Playwright, and CPU-scaled pytest workers

Booz Allen | Full-Stack Software Engineer Intern

Jun 2025 - Aug 2025

- Collaborated with two teammates to deliver a real-time cyber-physical analytics dashboard for a federal client, ingesting, processing, and visualizing live network activity from multiple ZeroTier-connected systems via an Elastic Stack pipeline; filtered over 99.9% of irrelevant traffic, improving clarity for non-technical users and reducing analyst review time
- Built a responsive React interface with D3-driven visualizations, reducing the time users needed to interpret traffic patterns and speeding up decision-making
- Implemented a data delivery layer by streaming real-time events via WebSockets and deployed optimized Express REST APIs backed by Redis, enabling rapid configuration and millisecond-level retrieval of live and historical data
- Prepared the application for production deployment by adding an Nginx reverse proxy for secure request handling, systemd service management for continuous uptime, and custom deployment scripts for cross-platform configuration, ensuring reliable operation in a live environment
- Presented the dashboard to a senior military general, receiving enthusiastic approval for performance and usability, securing support for near-term market release

Salisbury University | Computer Science Department Tutor & Lab Assistant

Aug 2024 - May 2026

- Provided conceptual and technical guidance on software engineering principles, artificial-intelligence techniques, and cybersecurity fundamentals to students at all levels, using Python and JavaScript examples and hands-on labs, which helped students grasp core concepts and complete assignments successfully
- Specialized in assisting students in introductory to advanced systems-programming courses on IPC, signals, threading, and data structures, demonstrating code in C on Linux and using debugging tools, which led to clearer understanding of concurrency concepts and higher project scores

Personal Projects

Lunara | Lunara

Oct 2025 - Dec 2025

- Won 1st place at Rutgers HealthHack 2025 for an AI-powered healthcare triage platform with sub-30-second routing
- Improved routing accuracy by nearly 20% across 140+ medical conditions while ensuring HIPAA compliance
- Developed NLP-based triage models trained on anonymized clinical call transcripts
- Built a full-stack system using React, FastAPI, and AWS with real-time confidence scoring
- Advanced the project under a Rutgers-affiliated completer program, presenting at a healthcare innovation conference presentation

Computer Vision Chick Counting

Dec 2025

- Developed an AI-based computer vision system for automated chick counting in high-throughput poultry settings
- Improved detection accuracy by approximately 2% through iterative model tuning, data refinement, and validation
- Prototyped real-time object detection pipelines using Ultralytics YOLO on Raspberry Pi hardware
- Industry-facing project developed with a commercial poultry partner; closed-source with patent consideration

Parallel Query Processing System | Parallel Query Processing System

Nov 2025 - Dec 2025

- Designed and implemented a distributed SQL-like query processing engine for large-scale data operations
- Parallelized query execution using MPI and OpenMP in C, achieving significant speedup over serial baselines
- Evaluated scalability through benchmarking of speedup, efficiency, and communication overhead

Skills

Languages: Python, JavaScript/TypeScript, C, C++, Go, Java, SQL, Bash, PHP, R, HTML & CSS

Web & Application Frameworks: React, Express, FastAPI, Flask, Gin, Tailwind, Bootstrap, Node, Bun, Tauri, D3.js, Plotly, Chart.js, Django, TanStack Query

Databases & Data Stores: MySQL, PostgreSQL, MongoDB, Redis, Neo4j (AuraDB), S3, RLS, JWT, Role-Based Access Control (RBAC)

Data Science & Data Processing: Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, ETL Pipelines, Ultralytics (YOLO), Jupyter, OpenCV, HuggingFace, Langchain, LLM fine-tuning, RAG

Cloud & Infrastructure: AWS, Docker, Containerization, VPS, Celery, Firebase, Nginx, Linux, WebSockets, MPI/OpenMP, Playwright

Tools & Practices: Git, Agile/Scrum, CI/CD Pipelines, Automated Testing, Vite, LaTeX, Figma, Cursor, Claude, Codex, Gemini API, Vim, Pytest